

Fundamentals and Applications of Spectroscopy

Course Code: CH 203 2025-26; Semester II

Instructors: Dr. Sairam S Mallajosyula and Dr. Saumyakanti Khatua

Offices: AB4/324 and AB6/335 [Office hour for this class will be decided after discussion with CR]

Email: msairam@iitgn.ac.in, khatuask@iitgn.ac.in

1.Mode of teaching: Offline

2.Venue: Please refer to the timetable for classroom.

3.Grading Policy:

- Exams: Overall weightage will be 50%. [Exam1: 25%, Exam2: 25%]
- Quizzes: Overall weightage will be 20%. There will be 2 quizzes. [Quiz1: before Exam-I, Quiz2: between Exam-I and exam-II]
- Group project with report: 15% [group projects will be given after Exam-I. The projects will involve hands-on measurements with various spectroscopic techniques that will be discussed in class. Each group will have different projects that will require similar level of engagement].
- Attendance: 15% [>80% attendance: 15%, 60%-80% attendance: 7.5%, 50-59% attendance : 5, <50% attendance: 0, taking attendance will be stopped after 15 minutes of the start of the class]

4.Course content with tentative schedule:4.1: The following topics will be covered before

Exam-I: Fundamentals quantum chemistry: Schrodinger's equation, basic description of wave functions, solution of particle in 1D box, Basic theoretical discussion on light-matter interaction from quantum mechanical approach, spectroscopic transitions and selection rules, Spectral broadening, rotational, vibrational (FTIR) spectroscopy.

4.2: The following topics will be covered between Exam-I and Exam-II: Introduction to NMR spectroscopy, chemical shifts, application of NMR and IR spectroscopy for determination of unknown chemicals, introduction to electronic spectroscopy. Introduction to fluorescence, fluorescence imaging and applications of fluorescence in biological systems. Introduction to Raman spectroscopy and its applications.

Additionally, following demo/hands-on measurements will be performed. These will be scheduled in sync with the lectures (and batch strength):

- (i) Measurement of FTIR spectrum of simple chemicals
- (ii) Determination of concentration of a pollutant gas [uses principles of vibrational spectroscopy]
- (iii) Measurement of NMR spectrum of a simple chemical
- (iv) Measurement of absorption and fluorescence spectrum of simple organic molecules

5.Suggested books:

- Fundamentals of molecular spectroscopy by Banwell and McCash, Mc Graw Hill, 4th edition
- Spectrometric Identification of Organic Compounds, 8ed (An Indian Adaptation) by Robert M. Silverstein, Wiley 8th edition.